

Formative assessment: data science mini project

Inspecting and manipulating data files on the command line

Formative overview

For the formative assessment you will be data wrangling and analyzing data from the NHANES cohort study.

You will work on this locally (on your computer) at first, and then after you have learnt to use high-performance computing on the University's BlueCrystal system, you will change to working there.

Problem-solving: A key skill for a data scientist is figuring out how to do something that you haven't done before. Throughout the unit, including in the formative and summative, you won't necessarily know all the commands needed and are expected to demonstrate your problem-solving skills by identifying solutions to particular tasks.

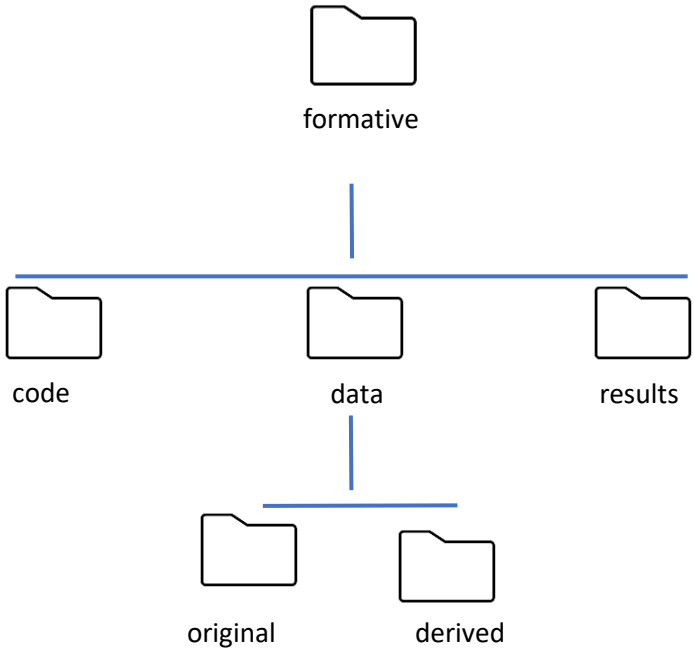
Often, there are multiple ways to complete a task, and it is up to you to decide which to use. The formative is a good opportunity to try different approaches to achieve the same result.

Section 1: Setting up directories and data files

You need somewhere to store your code, somewhere to store your results, and somewhere to store the data you'll work with.

It is important to get into good habits now, with a well-organised directory structure that you can build up as you work on the formative. It will help you (and others) to follow, and will make it less likely you will introduce bugs, e.g. getting confused about versions of datasets or results files.

Assuming that you already have a directory for this unit, make the directory structure shown here within that:



We will be using the **Physical activity monitor data** and the **body measures data** from the 2005-2006 NHANES cycle.

Physical activity monitor documentation – https://wwwn.cdc.gov/Nchs/Nhanes/2005-2006/PAXRAW_D.htm

Body measures documentation – https://wwwn.cdc.gov/Nchs/Nhanes/2005-2006/BMX_D.htm

Download the required datasets **from blackboard**. Extract the accelerometer files from the zip file (also called unzipping), using the command line.

Section 2. Exploring the data

Use the command line to explore the data. For example, find out:

- How many participants there are with body measures and accelerometer data
- What format are the data files in, and what information they contain

Make a text file in the code directory called README.txt. This file will contain information providing an overview to this project and the project code. Add a summary of the data to this text file.

Section 3. Processing the data

Perform some basic checks on the files using Linux commands, to check if there is anything awry with the data format. For example, you can check:

- all accelerometer files have the same header line
- there are the right number of columns on each row of each file

Write a bash script containing the commands you used to check the files.

Write a bash script to correct any issues you identify, so that the files are in a standard format (i.e. a header containing columns as the first row, and the same number of lines throughout).

Section 4. Generating a sample file

We want to understand the amount of data that might be usable, for our study that is looking at the relationship between physical activity and body measures.

We need to generate a list of participant IDs that will form our analytical sample. Your generated sample file should include two columns containing the participant IDs, and an indicator variable with values one or zero, indicating whether a participant is in our sample versus not in our sample.

Participants in our sample must have accelerometer data and a BMI measurement. All other participants are not in our sample.

You may wish to complete Section 4 using a combination of Linux commands and R, as appropriate.